EED 101 INTRODUCTION TO ELECTRICAL ENGINEERING

SPRING 2017 **INSTRUCTOR: DR G NAVEEN BABU**

PROBLEM SET 10

Reference text: Robert L. Boylestad, Louis Nashelsky, Electronic Devices and Circuit Theory, 11th Edition, Pearsoned

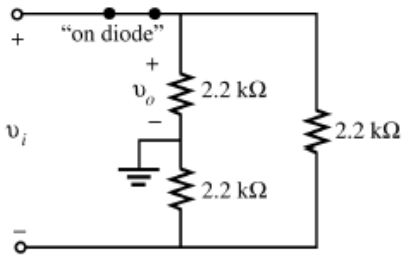
Refer the pdf extracted pages posted in blackboard.

Chapter 2: 30,31,32,33,34,35,36,37,38,39c,40,41

SOLUTIONS

30. Positive half-cycle of v_i:

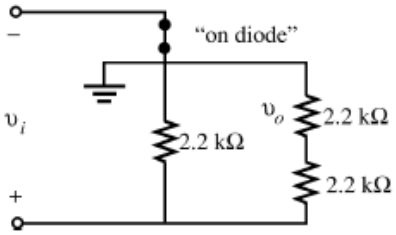
Network redrawn:



Voltage-divider rule:

$$V_{o_{max}} = \frac{2.2\text{ k}\Omega(V_{i_{max}})}{2.2\text{ k}\Omega + 2.2\text{ k}\Omega} = \frac{1}{2}(V_{i_{max}}) = \frac{1}{2}(100\text{ V}) = \mathbf{50\ V}$$

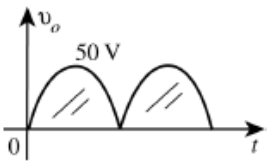
Negative half-cycle of v_i:



Polarity of v_o across the 2.2 kΩ resistor acting as a load is the same.

Voltage-divider rule:

$$V_{o_{max}} = \frac{2.2\text{ k}\Omega(V_{i_{max}})}{2.2\text{ k}\Omega + 2.2\text{ k}\Omega} = \frac{1}{2}(V_{i_{max}}) = \frac{1}{2}(100\text{ V}) = \mathbf{50\ V}$$



$V_{dc} = 0.636V_m = 0.636\ (50\text{ V}) = \mathbf{31.8\ V}$

31. Positive pulse of v_i:

Top left diode "off", bottom left diode "on"

2.2 kΩ || 2.2 kΩ = 1.1 kΩ

$$V_{o_{peak}} = \frac{1.1\text{ k}\Omega(170\text{ V})}{1.1\text{ k}\Omega + 2.2\text{ k}\Omega} = 56.67\text{ V}$$

Negative pulse of v_i:

Top left diode "on", bottom left diode "off"

$$V_{o_{peak}} = \frac{1.1\text{ k}\Omega(170\text{ V})}{1.1\text{ k}\Omega + 2.2\text{ k}\Omega} = 56.67\text{ V}$$

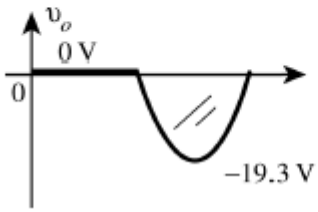
$V_{dc} = 0.636(56.67\text{ V}) = \mathbf{36.04\ V}$

32. (a) Si diode open for positive pulse of v_i and $v_o = \mathbf{0\ V}$

For $-20\text{ V} < v_i \le -0.7\text{ V}$ diode "on" and $v_o = v_i + 0.7\text{ V}$.

For $v_i = -20\text{ V}$, $v_o = -20\text{ V} + 0.7\text{ V} = \mathbf{-19.3\ V}$

For $v_i = -0.7\text{ V}$, $v_o = -0.7\text{ V} + 0.7\text{ V} = \mathbf{0\ V}$



(b) For $v_i \leq 8$ V the 8 V battery will ensure the diode is forward-biased and $v_o = v_i - 8$ V.

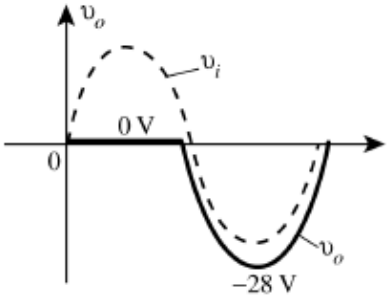
At $v_i = 8$ V

$v_o = 8\text{ V} - 8\text{ V} = \mathbf{0\ V}$

At $v_i = -20$ V

$v_o = -20\text{ V} - 8\text{ V} = \mathbf{-28\ V}$

For $v_i > 8$ V the diode is reverse-biased and $v_o = \mathbf{0\ V}$.

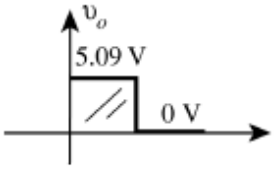


33. (a) Positive pulse of v_i:

$$V_o = \frac{1.8\text{ k}\Omega(12\text{ V} - 0.7\text{ V})}{1.8\text{ k}\Omega + 2.2\text{ k}\Omega} = \mathbf{5.09\ V}$$

Negative pulse of v_i:

diode "open", $v_o = \mathbf{0\ V}$

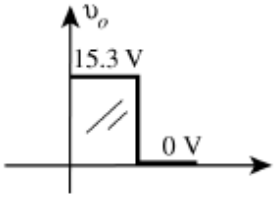


(b) Positive pulse of v_i:

$V_o = 12\text{ V} - 0.7\text{ V} + 4\text{ V} = \mathbf{15.3\ V}$

Negative pulse of v_i:

diode "open", $v_o = \mathbf{0\ V}$

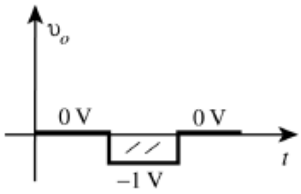


34. (a) For $v_i = 20$ V the diode is reverse-biased and $v_o = \mathbf{0\ V}$.

For $v_i = -5$ V, v_i overpowers the 4 V battery and the diode is "on".

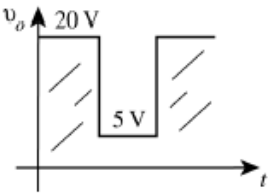
Applying Kirchhoff's voltage law in the clockwise direction:

$$-5\text{ V} + 4\text{ V} - v_o = 0$$
$$v_o = \mathbf{-1\ V}$$

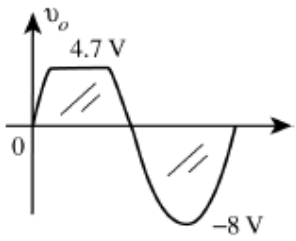


(b) For $v_i = 20$ V the 20 V level overpowers the 5 V supply and the diode is "on". Using the short-circuit equivalent for the diode we find $v_o = v_i = \mathbf{20\ V}$.

For $v_i = -5$ V, both v_i and the 5 V supply reverse-bias the diode and separate v_i from v_o. However, v_o is connected directly through the 2.2 kΩ resistor to the 5 V supply and $v_o = \mathbf{5\ V}$.



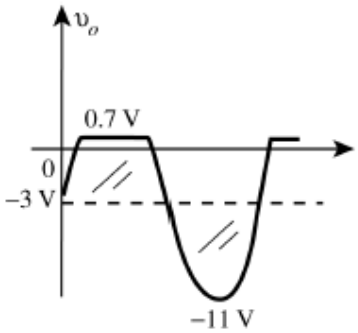
35. (a) Diode "on" for $v_i \geq 4.7$ V
For $v_i > 4.7$ V, $V_o = 4$ V + 0.7 V = **4.7 V**
For $v_i < 4.7$ V, diode "off" and $v_o = \boldsymbol{v_i}$



(b) Again, diode "on" for $v_i \geq 3.7$ V but v_o now defined as the voltage across the diode
For $v_i \geq 3.7$ V, $v_o =$ **0.7 V**

For $v_i < 3.7$ V, diode "off", $I_D = I_R = 0$ mA and $V_{2.2\,k\Omega} = IR = (0\text{ mA})R = 0$ V

Therefore, $v_o = v_i - 3$ V
At $v_i = 0$ V, $v_o =$ **−3 V**
$v_i = -8$ V, $v_o = -8$ V − 3 V = **−11 V**

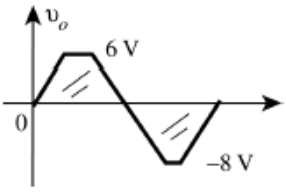


36. For the positive region of v_i:
The right Si diode is reverse-biased.
The left Si diode is "on" for levels of v_i greater than 5.3 V + 0.7 V = 6 V. In fact, $v_o =$ **6 V** for $v_i \geq 6$ V.

For $v_i < 6$ V both diodes are reverse-biased and $v_o = \boldsymbol{v_i}$.

For the negative region of v_i:
The left Si diode is reverse-biased.
The right Si diode is "on" for levels of v_i more negative than 7.3 V + 0.7 V = 8 V. In fact, $v_o =$ **−8 V** for $v_i \leq -8$ V.

For $v_i > -8$ V both diodes are reverse-biased and $v_o = \boldsymbol{v_i}$.



i_R: For −8 V < v_i < 6 V there is no conduction through the 10 kΩ resistor due to the lack of a complete circuit. Therefore, $i_R = 0$ mA.

For $v_i \geq 6$ V

$$v_R = v_i - v_o = v_i - 6 \text{ V}$$

For $v_i = 10$ V, $v_R = 10$ V − 6 V = 4 V

$$\text{and } i_R = \frac{4 \text{ V}}{10 \text{ k}\Omega} = \mathbf{0.4\ mA}$$

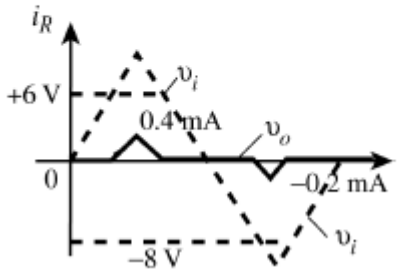
For $v_i \leq -8$ V

$$v_R = v_i - v_o = v_i + 8 \text{ V}$$

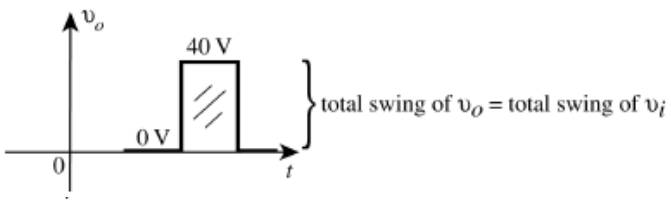
For $v_i = -10$ V

$$v_R = -10\text{ V} + 8\text{ V} = -2\text{ V}$$

$$\text{and } i_R = \frac{-2\text{ V}}{10\text{ k}\Omega} = \mathbf{-0.2\text{ mA}}$$

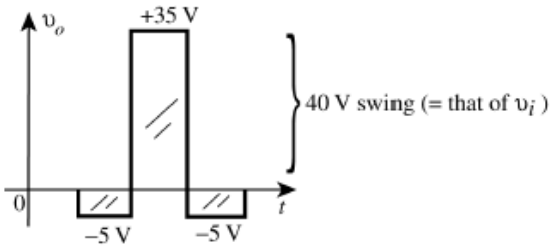


37. (a) Starting with $v_i = -20$ V, the diode is in the "on" state and the capacitor quickly charges to –20 V+. During this interval of time v_o is across the "on" diode (short-current equivalent) and $v_o = 0$ V.
When v_i switches to the +20 V level the diode enters the "off" state (open-circuit equivalent) and $v_o = v_i + v_C = 20\text{ V} + 20\text{ V} = +40\text{ V}$

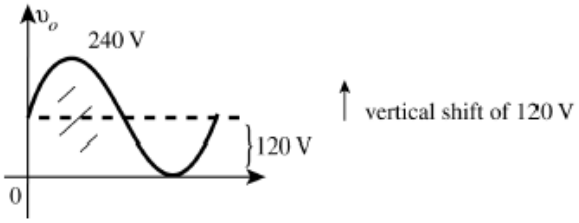


(b) Starting with $v_i = -20$ V, the diode is in the "on" state and the capacitor quickly charges up to –15 V+. Note that $v_i = +20$ V and the 5 V supply are additive across the capacitor. During this time interval v_o is across "on" diode and 5 V supply and $v_o = -5$ V.

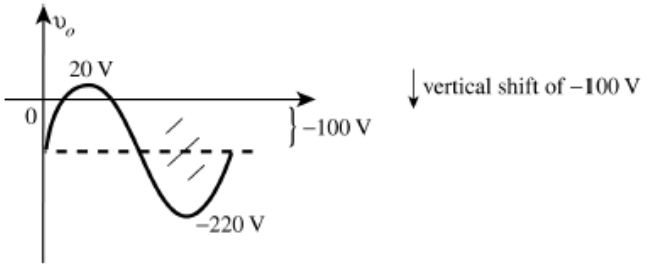
When v_i switches to the +20 V level the diode enters the "off" state and $v_o = v_i + v_C = 20\text{ V} + 15\text{ V} = 35\text{ V}$.



38. (a) For negative half cycle capacitor charges to peak value of 120 V = 120 V with polarity (– —||— +). The output v_o is directly across the "on" diode resulting in $v_o = \mathbf{0}$ **V** as a negative peak value.
For next positive half cycle $v_o = v_i + 120$ V with peak value of
$v_o = 120\text{ V} + 120\text{ V} = \mathbf{240\text{ V}}$.



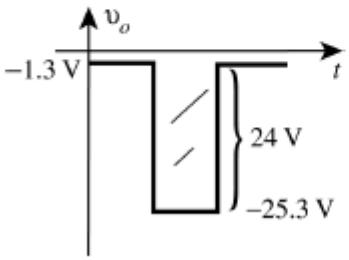
(b) For positive half cycle capacitor charges to peak value of 120 V – 20 V = 100 V with polarity (+ —||— –). The output v_o = 20 V = **20 V**
For next negative half cycle $v_o = v_i$ – 100 V with negative peak value of
v_o = –120 V – 100 V = **–220 V**.

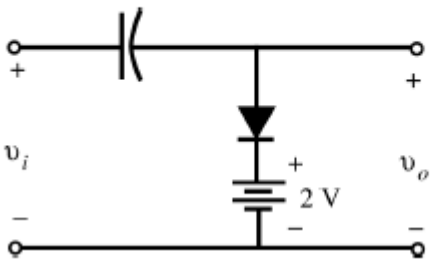


39.

(c) Positive pulse of v_i:
Diode "on" and v_o = –2 V + 0.7 V = –1.3 V
Capacitor charges to 12 V + 2 V – 0.7 V = 13.3 V

Negative pulse of v_i:
Diode "off" and v_o = –12 V – 13.3 V = –25.3 V



40. Solution is network of Fig. 2.181(b) using a 10 V supply in place of the 5 V source.

41. Network of Fig. 2.178 with 2 V battery reversed.